

# Instructions, Expressions and Operators in C++

## [DEF] Definition to Remember

Operators in C++ define computations on fundamental types and can be overloaded for user-defined classes. Expressions evaluate according to operator precedence and associativity.

## 1. Concept Overview & Fundamentals

Instructions, Expressions and Operators in C++ provides modern high-performance capabilities combining low-level hardware control with high-level abstractions, object-oriented paradigms, and Standard Template Library (STL) utilities.

In modern C++ (C++11 through C++20):

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## 2. Technical Architecture & Memory Mechanics

### C++ Runtime Execution Architecture



### Modern C++ Feature Comparison

| Modern C++ Paradigm                       | Legacy Equivalent                           | Engineering Advantage                  | Performance Impact                                  |
|-------------------------------------------|---------------------------------------------|----------------------------------------|-----------------------------------------------------|
| <code>**std::cin` / `std::cout`**</code>  | <code>`scanf` / `printf`</code>             | Type-safe, extensible for custom class | Buffered I/O (optimizable via <code>`ios::sy</code> |
| <code>**References (`Type&amp;`)**</code> | Pointers ( <code>`Type*`</code> )           | Non-nullable, clean syntax without der | Direct alias; zero runtime overhead                 |
| <code>**std::vector&lt;T&gt;`**</code>    | Raw Dynamic Array ( <code>`malloc`</code> ) | Automatic resizing, bounds safety, mem | Amortized <code>**O(1)**</code> push_back, contiguo |
| <code>**`nullptr`**</code>                | <code>`NULL` / `0`</code>                   | Strongly typed pointer literal prevent | Zero overhead                                       |

## 3. Core Syntax & Implementation Patterns

### Modern C++ Template Structure

```
#include <iostream>
#include <vector>
#include <string>
#include <algorithm>

// Fast I/O and standard namespace configuration
int main() {
    std::ios_base::sync_with_stdio(false);
    std::cin.tie(NULL);

    std::cout << "Executing C++ Module: Instructions, Expressions & Operators" << "\n";

    return 0;
}
```

## 4. Real-World Applications & System Engineering

- \* Quantitative Finance & HFT: High-frequency trading platforms use modern C++ for microsecond-level execution.

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## 5. Comprehensive Solved Problems & Code Walkthroughs

### Problem 1: AreaOfTriangle

Problem Description & Objective:

C++ Code Implementation:

Implem

```
/*
 * Que
 * Cre
 */

/*
 * ===
 * COM
 * ===
 * 1.
 *
 *
 * 2.
 *
 *
 * 3.
 *
 *
 * 4.
 * *
 *
 *
 * ===
 */

#include
using
```

```
//
int ma
//
do
do
do

//
co
//
ci

//
co
//
ci

//
an

//
co

//
re
}
```

## Problem 2: ArithmeticOperators

Problem Description & Objective:

C++ Code Implementation:

Implem

```
/*
 * Que
 * Cre
 */

/*
 * ==
 * CON
 * ==
 * 1.
 *
 *
 * 2.
 *
avoids
 *
 * 3.
 *
 *
 * 4.
 * *
 *
 * ==
 */

#include

using
```

```
//
int ma
//
in
in

//
co
//
ci

//
co
//
ci

//
co

//
co

//
if

//
}

}

//
re
}
```

### Problem 3: CompoundInterest

Problem Description & Objective:

C++ Code Implementation:

Implem

```
/*
 * Que
 * Cre
 */

/*
 * ==
 * CON
 * ==
 * 1.
```

```
*
*
* 2.
*
*
* 3.
*
*
* 4.
* *
*
*
*
* ==
*/

#include
#include

using

//
int main()
//
do
do
do
do
do
do

//
co
//
ci

//
co
//
ci

//
co
//
ci

//
am

//
co

//
co

//
re
}
```

## Problem 4: FahrenheitToCelsius

Problem Description & Objective:

C++ Code Implementation:

Implem

```
/*
 * Que
 * Cre
 * de
 */

/*
 * ==
 * CON
 * ==
 * 1.
 *
 *
 * 2.
 *
 *
 * 3.
 *
 *
 * 4.
 *
 *
 * ==
 */

#include

// Use
using

//
int ma
do

//
co
//
ci
ce

//
co

//
re
}

```

**Problem 5: IntToFloat**

Problem Description & Objective:

C++ Code Implementation:

Implem

```
/*
 * Que

```

```
* Giv
*/

/*
* ==
* CON
* ==
* 1
*
*
*
* 2.
*
*
* 3.
*
*
* 4.
*
*
*
* ==
*/

#include

// Use
using

//
int ma
in

//
co
//
ci

fl

//
co
//
co

//
re
}
```

---

### Problem 6: PerimeterRectangle

Problem Description & Objective:

C++ Code Implementation:

Implem

```
/*
* Que
* Cre
* Per
*/

/*
* ==
```

```
* CON
* ==
* 1.
*
*
* 2.
*
*
* 3.
*
*
* 4.
*
*
*
* ==
*/

#include

// Use
using

//
int ma
do

//
co
//
ci

//
co
//
ci

pe

//
co

//
re
}
```

## Problem 7: ProductFloat

Problem Description & Objective:

C++ Code Implementation:

Implem

```
/*
* Que
* Cre
*/

/*
* ==
* CON
* ==
* 1.
```

```
*
*
* 2.
*
*
* 3.
*
*
* 4.
*
*
*
* ===
*/

#include

// Use
using

//
int ma
fl

//
co
//
ci

//
co
//
ci

pr

//
co

//
re
}
```

## Problem 8: SimpleInterest

Problem Description & Objective:

C++ Code Implementation:

Implem

```
/*
* Que
* Cre
* Sim
*/

/*
* ===
* CON
* ===
* 1.
*
*
*
*/
```

```
* 2.
*
*
* 3.
*
*
* 4.
*
*
*
* ===
*/

#include

// Use
using

//
int ma
do

//
co
//
ci

//
co
//
ci

//
co
//
ci

si

//
co

//
re
}
```

## 6. Exam Scoring Points & Memory Revision

### [MUST-WRITE] Must-Write Points

1. Short-circuit evaluation: In `A && B`, `B` is not evaluated if `A` is false; in `A || B`, `B` is not evaluated if `A` is true.
2. Prefix vs Postfix increment: `++x` increments before evaluation; `x++` increments after evaluation.
3. Bitwise shifts: `x << k` multiplies by  $2^k$ ; `x >> k` divides by  $2^k$  for positive integers.
4. Ternary conditional operator: `condition ? expr1 : expr2` enables concise inline branch evaluations.
5. Explicit casting in modern C++: `static_cast<type>(expr)` ensures compile-time type safety.

### [RECALL] Quick Recall

Operator Precedence -> Short-Circuit Logic -> static\_cast -> Bitwise Manipulation -> Value Return